

Performance Testing

Date	08 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Testing
Type of Testing	Functional Testing & Performance Testing
Target Module / API	Loan Prediction Module
Test Environment	Local System (Flask Development Server)
Test Date	08-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid applicant details	1	5	Loan prediction displayed successfully
2	Submit invalid input values	1	5	Validation handled correctly
3	Multiple prediction requests	10	30	Application remains responsive

4	Load trained model and predict	5	20	Prediction generated without errors
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		12 req/sec	Pass	Good performance
4	Error Rate	< 1%	0%	Pass	No prediction errors
5	CPU Utilization	< 80%	42%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	48%	Pass	Normal memory consumption

Step 4: Observations & Analysis

Key Findings

The Smart Lender application successfully predicted loan approval with low response time and stable performance. All prediction requests were processed without failures.

Bottlenecks Identified

Minor delay occurs only during the initial loading of the trained model. No significant performance bottlenecks were observed.

Optimization Steps Taken

- Used a pre-trained Random Forest model.
- Loaded the model only once at application startup.
- Applied data scaling before prediction.
- Used Flask for lightweight request handling.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

